

Piecewise Linear Regression

Convex optimization problem:

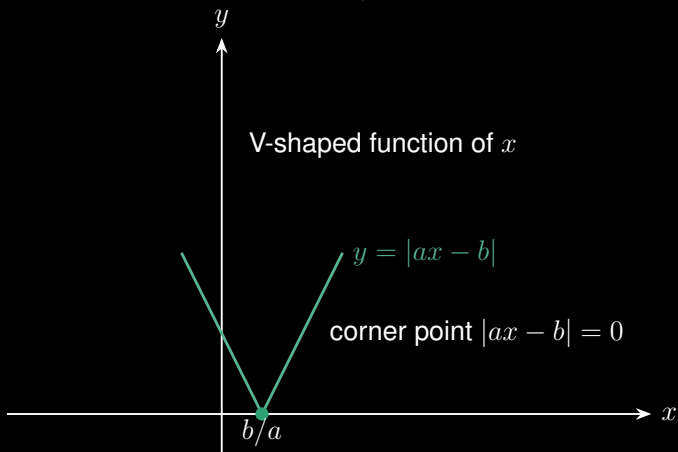
$$\min_{\mathbf{x}} f(\mathbf{x}) = \sum_{i=1}^n |a_i \mathbf{x} - b_i|$$

with data pairs $\{(a_i, b_i)\}_{i=1}^n$.

How to minimize $f(\mathbf{x})$ with a simple method?

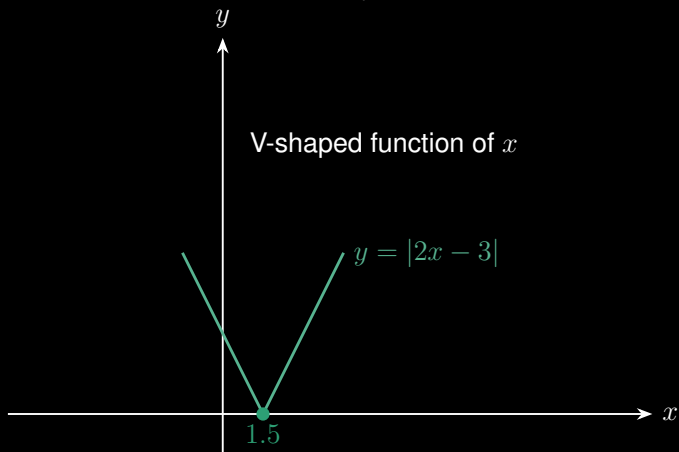
Piecewise Linear Regression

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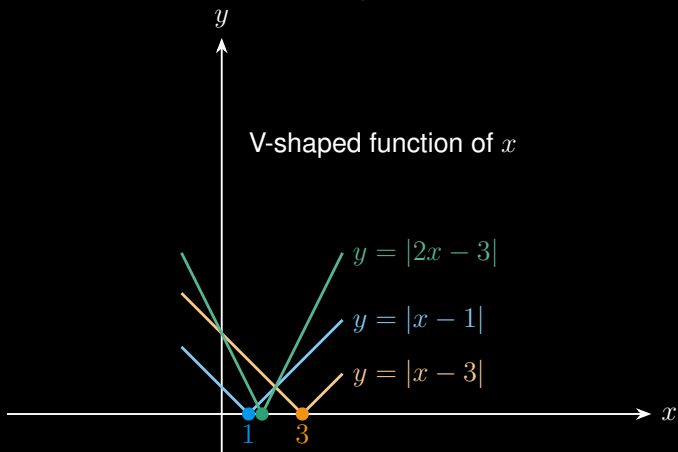
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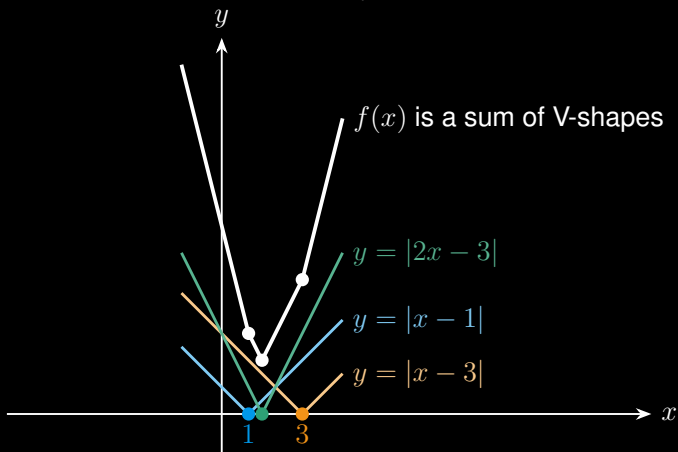
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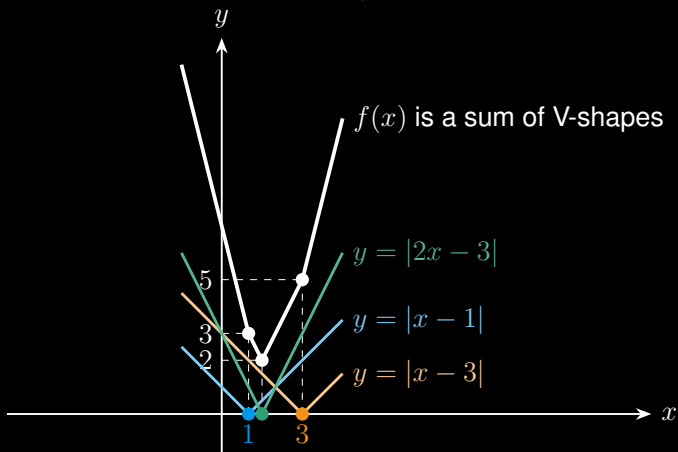
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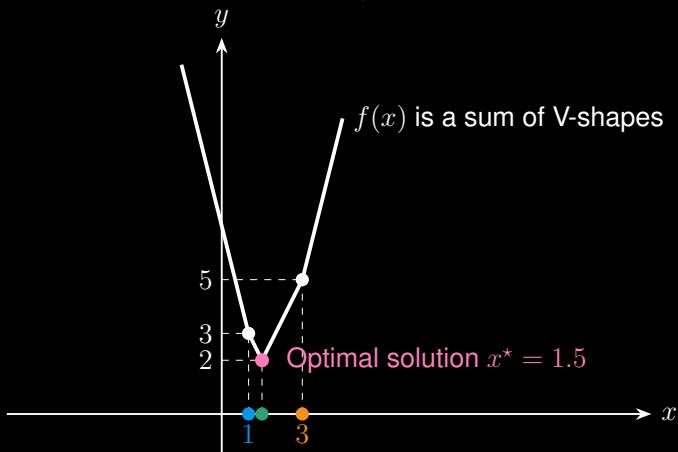
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Piecewise Linear Regression

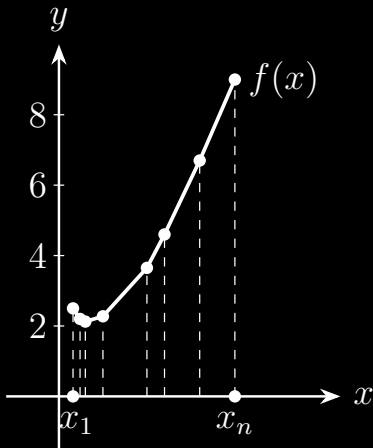
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Piecewise Linear Regression

$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i|$$

a_i	b_i	$x_i = b_i/a_i$
0.4	0.5	1.25
0.5	0.2	0.4
0.1	0.3	3
0.2	0.5	2.5
0.4	0.3	0.75
0.1	0.5	5
0.1	0.4	4
0.5	0.3	0.6

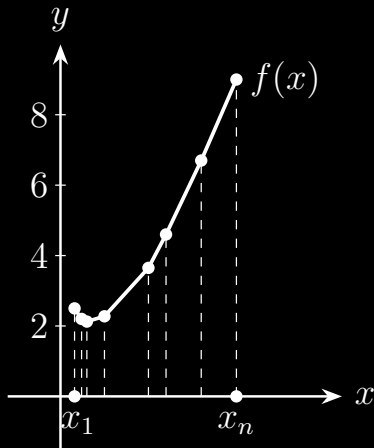


Piecewise Linear Regression

$$\min_{\mathbf{x}} f(\mathbf{x}) = \sum_{i=1}^n |a_i \mathbf{x} - b_i|$$

Sort values $x_1 \leq x_2 \leq \dots \leq x_n$:

$$x_i \in \{0.4, 0.6, 0.75, 1.25, 2.5, 3, 4, 5\}$$



Piecewise Linear Regression (Binary Search)

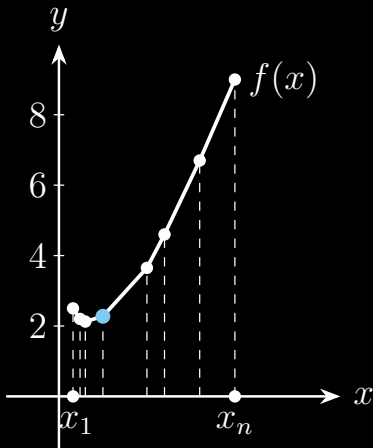
$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i|$$

Compute $k = \lfloor \frac{n}{2} \rfloor = 4$:

$$x_i \in \{0.4, 0.6, 0.75, \underbrace{1.25}_{x_4}, 2.5, 3, 4, 5\}$$

- Positive slope on $[x_4, x_5]$
- Positive slope on $[x_3, x_4]$

$f(x_4)$ is not the minimum value.



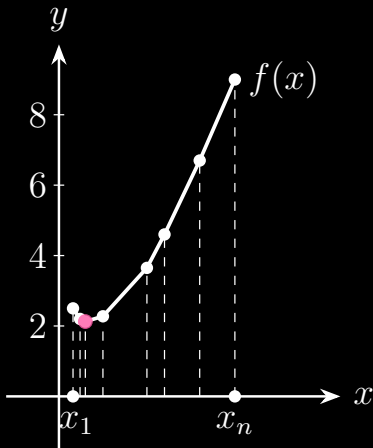
Piecewise Linear Regression (Binary Search)

$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i| \quad \text{where the optimal solution is } x^* = 0.75$$

$$x_i \in \{0.4, 0.6, \underbrace{0.75}_{x_3}, 1.25, 2.5, 3, 4, 5\}$$

- Positive slope on $[x_3, x_4]$
- Negative slope on $[x_2, x_3]$

$f(x_3)$ is the minimum value.

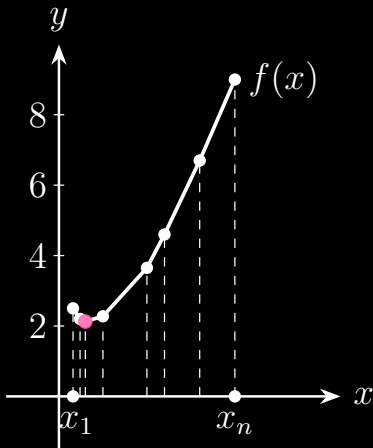


Piecewise Linear Regression

$$\min_{\mathbf{x}} f(\mathbf{x}) = \sum_{i=1}^n |a_i \mathbf{x} - b_i|$$

Algorithm:

- ① Compute all points $x_i = b_i/a_i$;
- ② Sort values $x_1 \leq x_2 \leq \dots \leq x_n$;



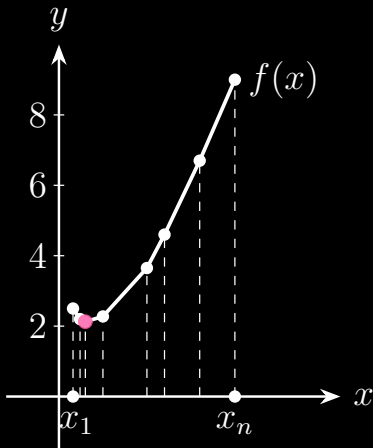
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- ① Compute all points $x_i = b_i/a_i$;
- ② Sort values $x_1 \leq x_2 \leq \dots \leq x_n$;
- ③ Let $k = \lfloor \frac{n}{2} \rfloor$ and compute the slope

$$s = \frac{f(x_{k+1}) - f(x_k)}{x_{k+1} - x_k}$$



Piecewise Linear Regression

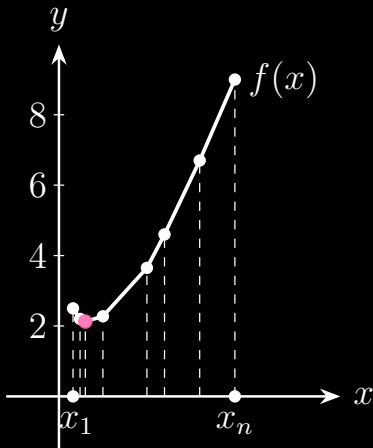
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$$s = \frac{f(x_{k+1}) - f(x_k)}{x_{k+1} - x_k}$$

- ④ Find some x_ℓ such that the slope is either zero on $[x_\ell, x_{\ell+1}]$ or negative on $[x_{\ell-1}, x_\ell]$ but positive on $[x_\ell, x_{\ell+1}]$.



Piecewise Linear Regression with Linear Programming

$$\min_{\mathbf{x}} f(\mathbf{x}) = \sum_{i=1}^n |a_i \mathbf{x} - b_i|$$

Linear programming problem:

$$\begin{aligned} \min_{\mathbf{x}, t_1, \dots, t_n} \quad & \sum_{i=1}^n t_i \\ \text{s.t.} \quad & -t_i \leq a_i \mathbf{x} - b_i \leq t_i \end{aligned}$$

CVXPY: Convex optimization
solver in Python.

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CVXPY: Convex optimization
solver in Python.

```
1 import cvxpy as cp
2
3 def piece_lr(a, b, n):
4     x = cp.Variable(1)
5     t = cp.Variable(n)
6
7     obj = cp.Minimize(cp.sum(t))
8     res = a * x - b
9     constraints = [-t <= res, res <= t]
10    prob = cp.Problem(obj, constraints)
11    result = prob.solve()
12
13    return x.value
```


Piecewise Linear Regression with Linear Programming

$$\min_x f(x) = \sum_{i=1}^n |a_i x - b_i| \quad \text{where the optimal solution is } x^* = 0.75$$

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0.4	0.5
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0.2	0.5
0.4	0.3
0.1	0.5
0.1	0.4
0.5	0.3

```
1 import numpy as np
2
3 n = 8
4 a = np.array([0.4, 0.5, 0.1, 0.2, 0.4, 0.1,
5               0.1, 0.5])
6 b = np.array([0.5, 0.2, 0.3, 0.5, 0.3, 0.5,
7               0.4, 0.3])
8
9 x = piece_lr(a, b, n)
10 print(x)
```

Thanks for your attention!

About me:

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 How to reach me: chenxy346@gmail.com